

A Jump Diffusion Model for Option Pricing

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Abstract

Brownian motion and normal distribution have been widely used in the Black-Scholes option pricing framework to model the return of assets. However, two puzzles emerge from many empirical investigations: the leptokurtic feature that the return distribution of assets may have a higher peak and two (asymmetric) heavier tails than those of the normal distribution, and an empirical abnormality called “volatility smile” in option pricing. To incorporate both of them, this paper proposes, for the purpose of option pricing, a double exponential jump diffusion model. The main attraction of the model is its simplicity. In particular, it is simple enough to derive analytical solutions for a variety of option pricing problems, including call and put options, interest rate derivatives and path-dependent options; it seems impossible for many other alternative models to do this. Equilibrium analysis and a psychological interpretation of the model are also presented.

Brownian motion and normal distribution have been widely used to study option pricing and the return of assets¹. Option pricing papers within the classical Black-Scholes model that are particularly relevant to the current paper are: Black and Scholes (1973) model for the call and put options; Black (1976) model for options on futures contracts; Heath, Jarrow and Morton (1992) model for options on bonds; and the market LIBOR models for swaptions, caps and floors, which are options on discretely compounded simple interest rates and are among the most liquid interest rate options (see Brace, Gatarek, and Musiela, 1997, Miltersen, Sandmann and Sondermann, 1997, and Jamshidian, 1997).

Despite the successes of the Black-Scholes model based on Brownian motion and normal distribution, two puzzles, emerged from many empirical investigations, have got much attention recently. (1) The asymmetric leptokurtic features. In other words, the return distribution is skewed to the left, and has a higher peak and two heavier tails than those of the normal distribution. (2) The volatility smile. More precisely, if the Black-Scholes-Merton model is correct, then the implied volatility should be constant; but it is widely recognized that the implied volatility curve resembles a “smile,” meaning it is a convex curve of the strike price.

Many researches have been conducted to modify the Black-Scholes model to explain the two puzzles. To incorporate the asymmetric leptokurtic features in asset pricing, a variety of models have been proposed, including, among others: (a) chaos theory, fractal Brownian

¹See, for example, Duffie (1995), Hull (1999), Ingersoll (1988), and Karatzas and Shreve (1998).

motion, and stable processes; see, for example, Mandelbrot (1963, 1967), Fama (1963, 1965), Rogers (1997), Samorodnitsky and Taqqu (1994); (b) generalized hyperbolic models, including log t model and log hyperbolic model; see, for example, Barndorff-Nielsen and Shephard (2001), Praetz (1972), Blattberg and Gonedes (1974); (c) time changed Brownian motions, including log variance gamma model; see, for example, Clark (1973), Madan and Seneta (1990), Madan, Carr, and Chang (1998), Andersen (1996), Hurst, Platen and Rachev (1997), Geman, Madan, and Yor (1998), and Heyde (2000). An immediate problem with these models is that it may be difficult to obtain analytical solutions for option pricing; more precisely, they might give some analytical formulae for regular call and put options, but certainly not for interest rate derivatives and path-dependent options, such as perpetual American options, barrier and lookback options.

In a parallel development, different models are also proposed to incorporate the “volatility smile” in option pricing. Popular ones are: (a) stochastic volatility and ARCH models; see, for example, Hull and White (1987), Engle (1995), Gouriéroux (1997), Fouque, Papanicolaou, and Sircar (2000); (b) constant elasticity model (CEV) model proposed by Cox and Ross (1976); (c) normal jump models proposed by Merton (1976); (d) affine stochastic-volatility and affine jump diffusion models; see, for example, Heston (1993) and Duffie, Pan, and Singleton (2000); (e) Models based on Lévy processes; see, for example, Geman, Madan, and Yor (1999) and reference therein; (f) a numerical procedure called “implied binomial trees”; see, for example, Derman and Kani (1994), Dupire (1994), Rubinstein (1994). Aside from the problem that it might not be easy to find analytical solutions for option pricing, especially for path-dependent options (such as perpetual American options, barrier and lookback options), some of these models may not produce the asymmetric leptokurtic feature; see Section I.C.

The current paper proposes a new model, which has the following properties. (a) It offers an explanation to two empirical phenomena: the asymmetric leptokurtic feature, and the volatility smile; see Sections II and V. (b) It leads to analytical solutions to many option pricing problems, including European call and put options (see Section IV); interest rate derivatives, such as swaptions, caps, floors, and bond options (see Section V and Glasserman and Kou, 1999); path-dependent options, such as perpetual American options, barrier and lookback options (see Section I.C and Kou and Wang, 2000, 2001). (c) It can be embedded into a rational expectation equilibrium framework (see Section III). (d) It has a psychological interpretation (see Section I.B).

The model is very simple. It consists of two parts, a continuous part driven by a geometric Brownian motion, and a jump part, with the logarithm of jump sizes having a double expo-

nential distribution and the jump times corresponding to the event times of a Poisson process. Because of its simplicity, the parameters in the model can be easily interpreted, and the analytical solutions for option pricing can be obtained. The explicit calculation is made possible partly because of the memoryless property of the double exponential distribution and a special function, the Hh function.

The paper is organized in the following way. In Section I, the model is proposed, is evaluated by five criteria, and is compared with other alternative models. Section II studies the leptokurtic feature. A rational expectation equilibrium justification of the model is given in Section III. Some preliminary results, including the Hh functions, are given in Section IV.A. Formulae for option pricing problems, including options on futures, are provided in Section IV.B. Section V.A studies the pricing of interest rate options, such as caps and bond options. “Volatility smiles” is illustrated in Section V.B. The last section discusses the advantages and disadvantages of the model.

I The Model

I.A The Formulation of the Model

The model that I propose for the price of an underlying asset (for example a stock or a stock index) consists of two parts, a continuous part driven by a geometric Brownian motion, and a jump part, with the logarithm of jump sizes having a double exponential distribution and the jump times corresponding to the event times of a Poisson process. More precisely, the following stochastic differential equation is used to model the asset price, $S(t)$,

$$\frac{dS(t)}{S(t-)} = \mu dt + \sigma dW(t) + d \left(\sum_{i=1}^{N(t)} (V_i - 1) \right), \quad (1)$$

where $W(t)$ is a standard Brownian motion, $N(t)$ a Poisson process with rate λ , and $\{V_i\}$ a sequence of independent identically distributed (i.i.d.) nonnegative random variables such that $Y = \log(V)$ has an asymmetric double exponential distribution^{2,3,4} with the density

$$f_Y(y) = p \cdot \eta_1 e^{-\eta_1 y} 1_{\{y \geq 0\}} + q \cdot \eta_2 e^{\eta_2 y} 1_{\{y < 0\}}, \quad \eta_1 > 1, \eta_2 > 0,$$

²If $\eta_1 = \eta_2$ and $p = 1/2$, then the double exponential distribution is also called “the first law of Laplace” (proposed by Laplace in 1774), while the “second law of Laplace” is the normal density.

³Although at the point of the mode, the (asymmetric) double exponential density has a discontinuity in the derivative, it will be smoothed out by the normal distribution from the continuous Brownian motion part, resulting in a density everywhere differentiable for the process in (1).

⁴Ramezani and Zeng (1998) independently propose the same jump diffusion model from an econometric viewpoint in order to improve the empirical fit of Merton’s normal jump diffusion model to stock price data.

where $p, q \geq 0$, $p + q = 1$, represent the probabilities of upward and downward jumps. In other words,

$$\log(V) = Y \stackrel{d}{=} \begin{cases} \xi^+, & \text{with probability } p \\ -\xi^-, & \text{with probability } q \end{cases}, \quad (2)$$

where ξ^+ and ξ^- are exponential random variables with means $1/\eta_1$ and $1/\eta_2$, respectively, and the notation $\stackrel{d}{=}$ means equal in distribution. In the model, all sources of randomness, $N(t)$, $W(t)$, and Y 's, are assumed to be independent, although this can be relaxed, as will be suggested in Section I.B. For notational simplicity, and in order to get analytical solutions for various option pricing problems, the drift μ and the volatility σ are assumed to be constants, and the Brownian motion and jumps are assumed to be one-dimensional. These assumptions, however, can be easily dropped to develop a general theory.

Solving the stochastic differential equation (1) gives the dynamics of the asset price:

$$S(t) = S(0) \exp \left\{ \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W(t) \right\} \prod_{i=1}^{N(t)} V_i. \quad (3)$$

Note that $E(Y) = \frac{p}{\eta_1} - \frac{q}{\eta_2}$, $\text{Var}(Y) = pq \left(\frac{1}{\eta_1} - \frac{1}{\eta_2} \right)^2 + \left(\frac{p}{\eta_1^2} + \frac{q}{\eta_2^2} \right)$, and

$$E(V) = E(e^Y) = q \frac{\eta_2}{\eta_2 + 1} + p \frac{\eta_1}{\eta_1 - 1}, \quad \eta_1 > 1, \eta_2 > 0. \quad (4)$$

The requirement $\eta_1 > 1$ is needed to ensure that $E(V) < \infty$ and $E(S(t)) < \infty$; it essentially means that the average upward jump cannot exceed 100%, which is quite reasonable.

There are two interesting properties of the double exponential distribution, which are crucial for the model. First, it has the leptokurtic feature; see, for example, Kotz, Johnson, and Balakrishnan (1995). As will be shown in Section II, the leptokurtic feature of the jump size distribution leads to the same feature for the return distribution. Secondly, a unique feature, inherited from the exponential distribution, of the double exponential distribution is the memoryless property. This special property explains why the closed form solutions for various option pricing problems, including barrier, lookback, and perpetual American options, are feasible under the double exponential jump diffusion model, while it seems impossible for many other model, e.g. the normal jump diffusion model (Merton, 1976); see Section I.C for details.

I.B Evaluating the Model

Since essentially all models are wrong, instead of arguing the correctness of the model I shall evaluate and justify the double exponential jump diffusion model by five criteria.

(1) A model must be self-consistent. In finance context, it means that a model must be arbitrage-free and can be embedded in an equilibrium setting. Note that some of alternative models may have arbitrage opportunities, thus are not self-consistent; for example, it is shown in Rogers (1997) that models using fractal Brownian motion may lead to arbitrage opportunities. The double exponential jump diffusion model can be embedded in a rational expectation equilibrium setting; see Section III.

(2) A model must be able to explain some empirical phenomena. The double exponential jump diffusion model is able to reproduce the leptokurtic feature for the return of assets (see Section II), and the “volatility smile” observed in option pricing (see Section V.B).

(3) A model must be simple enough to be computed. Like the Black-Scholes model, the double exponential jump diffusion model not only yields closed form solutions for call and put options (see Section IV), but also leads to a variety of closed form solutions, including path-dependent options, such as barrier options, lookback options, perpetual American options (see Section I.C and Kou and Wang, 2000, 2001), and interest rate derivatives (see Section V and Glasserman and Kou, 1999).

(4) A model must have some (physical, psychological, etc.) interpretation. One motivation of the double exponential jump diffusion model comes from behavioral finance. It has been suggested from extensive empirical studies that markets tend to have *both overreaction and underreaction* to various good or bad news (instead of giving a long list of references, the reader is referred to, for example, the papers by Fama, 1998, and Barberis, Shleifer, and Vishny, 1998, and references therein).

One may interpreted the jump part in the model as the market responses to outside news. More precisely, good or bad news arrives according to a Poisson process, and the asset price changes afterwards according to the jump size distribution. Because the double exponential distribution⁵ has both a high peak and heavy tails, it can be used to model both the overreaction (attributed to the heavy tails) and underreaction (attributed to the high peak) to outside news. Therefore, the double exponential jump diffusion model can be interpreted as an attempt to build a simple model, within the traditional random walk and efficient market framework, to incorporate investors’ sentiment.

Incidentally, as a by product, the model also suggests that the fact of markets having both overreaction and underreaction to outside news can lead to the leptokurtic feature of asset

⁵Interesting enough, the double exponential distribution has been widely used in mathematical psychology literature, particularly vision cognitive studies; see, for example, the list of papers on the web page of David Mumford at the computer vision group, Brown University.

return distribution.

(5) A model must be simple enough to be extended, if necessary. Since essentially all the models are “wrong”, it is desirable to have models with the flexibility of being modified easily to better fit the reality. One empirical observation that the double exponential jump diffusion model cannot incorporate is the possible dependence structure among asset returns (so-called the “volatility clustering effect”), simply because the double exponential jump diffusion model has independent increments. However, an easy way to incorporate the possible dependent structure is to use some other point process, $\tilde{N}(t)$, with dependent increments, to replace the Poisson process $N(t)$, while still retaining the independence between the Brownian motion, the jump sizes, and $\tilde{N}(t)$. In doing so, the modified model no longer exhibits independent increments, yet is simple enough to produce closed form solutions, at least for call and put options. However, it seems impossible to get analytical solutions for path-dependent options by using the new process $\tilde{N}(t)$ instead of $N(t)$.

I.C Comparison with Other Models

There are many alternative models that can satisfy at least some of the five criteria listed above. So what is interesting here? The main attraction of the double exponential jump diffusion model is its *simplicity, particularly its analytical tractability for path-dependent options and interest rate derivatives*.

More precisely, unlike the original Black-Scholes model, many alternative models can only compute prices for call and put options, and cannot derive analytical solutions for other equity derivatives, such as path-dependent options and for some most liquidate interest rate derivatives, such as swaption, caps, and floors. Even numerical methods for interest rate derivatives and path-dependent options are not easy. For example, the convergence rates of binomial trees and Monte Carlo simulation for path-dependent options are typically much slower than those for call and put options; for a survey, see, e.g., Boyle, Broadie, Glasserman (1997).

This makes it harder to persuade people to switch from the Black-Scholes model to alternative models. The double exponential jump diffusion model attempts to improve the empirical implication of the Black-Scholes model, while still retaining its analytical tractability. Below is a more detailed comparison between the current model and some popular alternative models.

(1) The CEV Model. Like the double exponential jump diffusion model, analytical solutions for path-dependent options (see Davydov and Linetsky, 1999) and interest rate derivatives (e.g. Cox, Ingersoll and Ross, 1985, and Andersen and Andersen, 2000) are available under the

CEV model. However, the CEV model does not have the leptokurtic feature; more precisely, the return distribution in the CEV model has a thinner right tail than that of the normal distribution. This undesirable feature also has a consequence in terms of the implied volatility in option pricing; more precisely, under the CEV model the implied volatility can only be a monotone function of the strike price. Therefore, if the implied volatility is a convex function (but not necessarily a decreasing function), as frequently observed in option markets, the CEV is unable to fit the implied volatility curve. For other surprising behavior of the CEV model, see Heath and Schweizer (2000).

(2) The normal jump diffusion model. Merton (1976) first considered the jump diffusion model similar to (1) and (3). In that paper Y 's are assumed to have normal distribution rather than the double exponential distribution. Both the double exponential and normal jump diffusion models can lead to the leptokurtic feature (although the kurtosis from the double exponential jump diffusion model is bigger), implied volatility smile, and analytical solutions for call and put options, and interest rate derivatives (such as caps, floors, and swaptions; see Glasserman and Kou, 1999). The main difference between the double exponential jump diffusion model and the normal jump diffusion model is the analytical tractability for the path-dependent options, such as barrier, lookback, and perpetual American options.

Here I provide some intuition to understand why the double exponential jump diffusion model can lead to closed form solutions for path-dependent options, while the normal jump diffusion model cannot. To price perpetual American options, barrier options, and lookback options for general jump diffusion processes, it is crucial to study the first passage times that a jump diffusion process crosses a flat boundary. When a jump diffusion process crosses boundary level sometimes it hits the boundary exactly and sometimes it incurs an “overshoot” over the boundary.

The overshoot presents several problems, if one wants to compute the option prices explicitly. First, one needs to get the exact distribution of the overshoot. It is well known from stochastic renewal theory that this is only possible if the jump size Y has an exponential type distribution, thanks to the special memoryless property of the exponential distribution. Secondly, one needs to know the dependent structure between the overshoot and the first passage time that the process crosses the barrier. The two random variables are conditionally independent, given that the overshoot is bigger than 0, if the jump size Y has an exponential type distribution, again thanks to the memoryless property. This conditionally independent structure seems to be very special to the exponential type distribution, and does not hold for other distributions,

such as the normal distribution.

Consequently, analytical solutions⁶ of the perpetual American, lookback, and barrier options can be derived for the double exponential jump diffusion model. However, it seems impossible to get similar results for other jump diffusion processes, e.g. the normal jump diffusion model.

(3) Models based on t -distribution. t -distribution is widely used in empirical studies of asset pricing. A problem of using t -distribution (or other distributions with power type tails) as return distribution is that it cannot be used in continuously compounded models. More precisely, suppose that, at time 0, the daily return distribution X has a power type right tail. Then in the continuously compounded finance models, the asset price tomorrow $A(\Delta t)$ is given by $A(\Delta t) = A(0) \cdot e^X$. Since X has a power type right tail, it is clear that $E(e^X) = \infty$. Consequently,

$$E(A(\Delta t)) = E(A(0)e^X) = A(0)E(e^X) = \infty.$$

In other word, the asset price tomorrow has an infinite expectation! This paradox holds for t -distribution with any degrees of freedom, as long as one considers the continuously compounded models. Furthermore, if the risk-neutral return also has a power type right tail, then the call option in the continuously compounded models is also infinite:

$$E^*((A(\Delta t) - K)^+) \geq E^*(A(\Delta t) - K) = \infty.$$

Therefore, the only relevant models in which returns have t -distributions are models with discretely compounded return. In discrete compounded models, closed form solutions, however, are in general impossible⁷.

(4) Stochastic volatility models. The double exponential jump diffusion model and the stochastic volatility model complement each other: the stochastic volatility model can incorporate dependent structure better, and the double exponential jump diffusion model has better

⁶The closed form solutions for the barrier and lookback options are given in terms of the Laplace transforms. This is mainly because the correlation between the terminal value of the asset at expiration and the overshoot is not known, even for jump processes; see Kou and Wang (2000) for details.

⁷Another interesting point worth of mentioning is that, for a sample size of 5000 (20-years-daily data), it may be very difficult to distinguish the double exponential distribution from the power type distributions, such as t -distribution; for example, the histograms of the two distributions are almost indifferent to human eyes. In fact, a forthcoming paper by C.C. Heyde and myself points out that many statistical procedures may fail to detect the differences between exponential type tails and power type tails, mainly because it requires a very large number of data to distinguish the two (although it is quite easy to detect the differences between them and the tails of the normal density). It should be emphasized, however, that here I am not arguing which one is more suitable to fit empirical observations. What is suggested here is that, since the t -distribution cannot in general give closed form solutions (because essentially they have to deal with discretely compounded models), the double exponential distribution may be a good alternative choice, as a balance between reality and analytical tractability.

analytical tractability, especially for path-dependent options and complex interest rate derivatives.

One empirical phenomenon worth mentioning is that the daily return distribution tends to have more kurtosis than the distribution of monthly returns. As Das and Foresi (1996) points out, this is consistent with models with jumps, but inconsistent with stochastic volatility models. More precisely, in stochastic volatility models (or essentially any models in a pure diffusion setting) the kurtosis decreases as the sampling frequency increases; while in jump models the instantaneous jumps are independent of the sampling frequency.

Of course, one can think of combining the stochastic volatility model with jump models, which will lead to the next class of models:

(5) Affine jump diffusion models. Duffie, Pan, and Singleton (2001) proposes a very general class of affine jump diffusion models which can incorporate jumps, stochastic volatility, and jumps in volatility. Both normal and double exponential jump diffusion models can be viewed as special cases of their model. However, because of the special features of the exponential distribution, the double exponential jump diffusion model can lead to analytical solutions for path-dependent options, which are difficult for other affine jump diffusion models. In particular, it seems quite difficult to get analytical solutions for path-dependent option if one combines the double exponential jump diffusion and stochastic volatility. Furthermore, the double exponential model is simpler than general affine jump diffusion models; namely it has fewer parameters, which make calibration easier, and pricing formulae (e.g. those for call and put options, caps, floors, swaptions) do not involve the Fourier inversion.

(6) Models based on Lévy processes (meaning the processes with independent and stationary increments). Although the double exponential jump diffusion model is a special case of Lévy processes, because of the special features of the exponential distribution, it has a good analytical tractability for path-dependent options and interest rate derivatives, which are difficult for other Lévy processes.

II Leptokurtic Feature

Using (3), the return over time Δt is given by

$$\frac{\Delta S(t)}{S(t)} = \frac{S(t + \Delta t)}{S(t)} - 1 = \exp \left\{ \left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma (W(t + \Delta t) - W(t)) + \sum_{i=N(t)+1}^{N(t+\Delta t)} Y_i \right\} - 1,$$

where a summation over an empty set is taken to be zero. If the time interval Δt is small, as in the case of daily observations, the return can be approximated, ignoring the terms with order higher than Δt and using the expansion $e^x \approx 1 + x + x^2/2$, in distribution by

$$\frac{\Delta S(t)}{S(t)} \approx \mu \Delta t + \sigma Z \sqrt{\Delta t} + B \cdot Y, \quad (5)$$

where B is a Bernoulli random variable with $P(B = 1) = \lambda \Delta t$ and $P(B = 0) = 1 - \lambda \Delta t$, and Y is given by (2).

By using (35) and (36) in Appendix B, of the density of the sum of a double exponential and a normal random variable, the density g of the right hand side of (5), being an approximation for the return $\frac{\Delta S(t)}{S(t)}$, is given by

$$\begin{aligned} g(x) = & \lambda \Delta t \{ p \eta_1 e^{(\sigma^2 \eta_1^2 \Delta t)/2} e^{-(x - \mu \Delta t) \eta_1} \Phi \left(\frac{x - \mu \Delta t - \sigma^2 \eta_1 \Delta t}{\sigma \sqrt{\Delta t}} \right) \\ & + q \eta_2 e^{(\sigma^2 \eta_2^2 \Delta t)/2} e^{(x - \mu \Delta t) \eta_2} \Phi \left(-\frac{x - \mu \Delta t + \sigma^2 \eta_2 \Delta t}{\sigma \sqrt{\Delta t}} \right) \} + \frac{1 - \lambda \Delta t}{\sigma \sqrt{\Delta t}} \varphi \left(\frac{x - \mu \Delta t}{\sigma \sqrt{\Delta t}} \right), \end{aligned}$$

where $\varphi(\cdot)$ is the standard normal density function. The density g has the mean and the variance given by

$$E_g(G) = \mu \Delta t + \lambda \left(\frac{p}{\eta_1} - \frac{q}{\eta_2} \right) \Delta t \quad (6)$$

$$\text{Var}_g(G) = \sigma^2 \Delta t + \left\{ pq \left(\frac{1}{\eta_1} + \frac{1}{\eta_2} \right)^2 + \left(\frac{p}{\eta_1^2} + \frac{q}{\eta_2^2} \right) \right\} \lambda \Delta t + \left(\frac{p}{\eta_1} - \frac{q}{\eta_2} \right)^2 \lambda \Delta t (1 - \lambda \Delta t). \quad (7)$$

An important feature of this asset return density is that, comparing to the normal density with the identical mean and variance, it has a higher peak around the mean, and two heavier tails (could be asymmetric); or, in short, the asymmetric leptokurtic feature.

Below are figures of the density, $g(x)$, compared with the normal density with the same mean and variance given by (6) and (7). The first figure compares the overall shapes of the two densities, the second one details the shapes around the peak areas, and the last two show the left and right tails. The dot line is used for the normal density, and the solid line is used for the model. The parameters used here are $\Delta t = 1 \text{ day} = 1/250 \text{ year}$, $\sigma = 20\%$ per year, $\mu = 15\%$ per year, $\lambda = 10$ per year, $p = 0.30$, $1/\eta_1 = 2\%$, and $1/\eta_2 = 4\%$. In this case, $E(Y) = -2.2\%$, and $\text{SD}(Y) = 4.47\%$. In other words, there are about 10 jumps per year with average jump size -2.2% , and the jump volatility 4.47% . The jump parameters used here seem to be quite reasonable, if not conservative, for a U.S. stock.

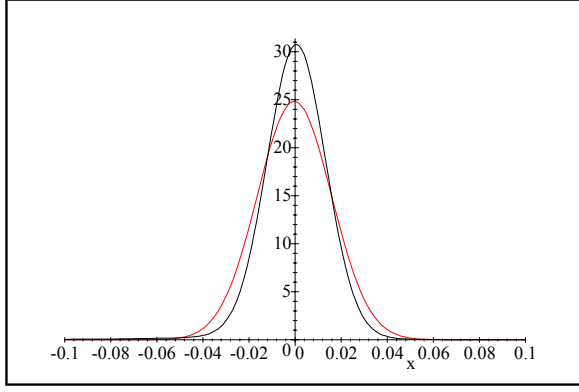


Figure 1: Overall Comparison

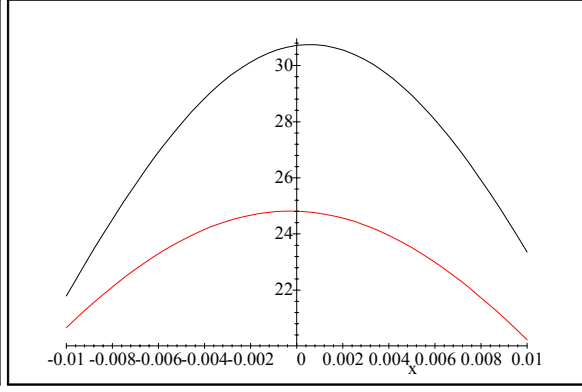


Figure 2: Peak Comparison

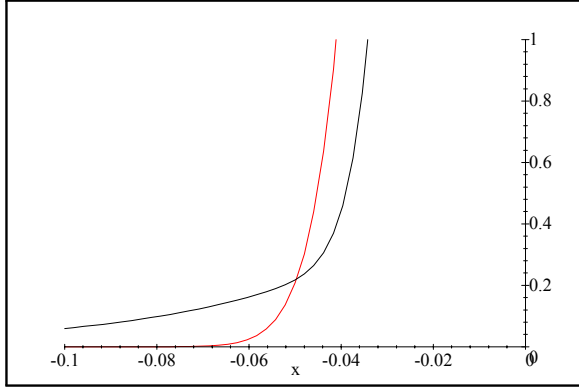


Figure 3: Left Tail Comparison

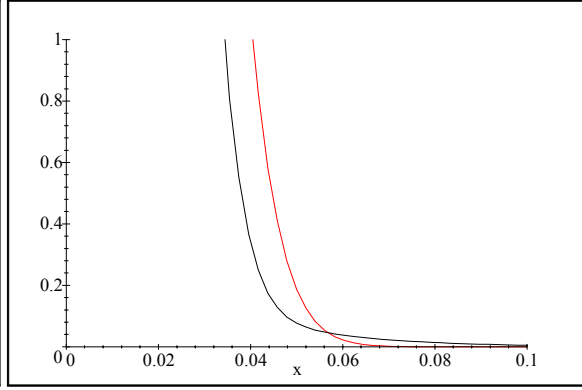


Figure 4: Right Tail Comparison

The leptokurtic feature is quite evident. The peak of the density g is about 31, whereas that of the normal density is about 25. The density g has heavier tails than the normal density, especially for the left tail, which could reach well below -10% , while the normal density is basically confined within -6% . Additional numerical plots suggest that the feature of higher peak and heavier tails becomes more significant if either $1/\eta_i$ (the jump size expectations) or λ (the jump rate) increases.

III Equilibrium for General Jump Diffusion Models

Consider a typical rational expectation economy (Lucas, 1978) in which a representative investor tries to solve a utility maximization problem

$$\max_c \mathbb{E} \left[\int_0^\infty U(c(t), t) dt \right],$$

where $U(c(t), t)$ is the utility function of the consumption process $c(t)$. There is an exogenous endowment process, denoted by $\delta(t)$, available to the investor. Also given to the investor is

an opportunity to invest in an asset which pays no dividends. If $\delta(t)$ is Markovian, it can be shown (see, for example, pp. 484-485 in Stokey and Lucas, 1989) that, under mild conditions, the rational expectation equilibrium price (also called the “shadow” price) of the asset, $p(t)$, must satisfy the Euler equation

$$p(t) = \frac{E(U_c(\delta(T), T)p(T)|\mathcal{F}_t)}{U_c(\delta(t), t)}, \quad \forall T \geq t \geq 0, \quad (8)$$

where U_c is the partial derivative of U with respect to c . At this price $p(t)$, the investor will never invest in (either long or short) the asset, though he/she is given the opportunity to do so. Instead, in equilibrium the investor find it optimal to just consume the exogenous endowment, i.e. $c(t) = \delta(t)$, for all $t \geq 0$.

In this section, I shall derive explicitly the implication of the Euler equation (8) when the endowment process $\delta(t)$ follows a general jump diffusion process:

$$\frac{d\delta(t)}{\delta(t-)} = \mu_1 dt + \sigma_1 dW_1(t) + d \left[\sum_{i=1}^{N(t)} (\tilde{V}_i - 1) \right], \quad (9)$$

where $\tilde{V}_i \geq 0$ can be any independent identically distributed, nonnegative random variables. In addition, all three sources of the randomness, the Poisson process $N(t)$, the standard Brownian motion $W_1(t)$, and the jump sizes $\tilde{V}$ are all assumed to be independent.

The work in this section builds upon and extends the previous work by Naik and Lee (1990), in which the special case that $\tilde{V}_i$ has a lognormal distribution is investigated. Another difference is that Naik and Lee (1990) requires that the asset pays continuous dividends, and there is no outside endowment process; while here the asset pays no dividend, and there is an outside endowment process. Consequently, the pricing formulae are different even in the case of lognormal jumps.

For simplicity, as in Naik and Lee (1990), I shall only consider the utility function of the special forms

$$U(c, t) = e^{-\theta t} \frac{c^\alpha}{\alpha}, \quad 0 < \alpha < 1, \quad \text{and} \quad U(c, t) = e^{-\theta t} \log(c), \quad \text{if } \alpha = 0,$$

where $\theta > 0$, although most of the results below hold for more general utility functions. Under these types of utility functions, the rational expectation equilibrium price of (8) becomes

$$p(t) = \frac{E(e^{-\theta T} (\delta(T))^{\alpha-1} p(T) | \mathcal{F}_t)}{e^{-\theta t} (\delta(t))^{\alpha-1}}. \quad (10)$$

Assumption. The discount rate θ should be large enough such that

$$\theta > -(1 - \alpha)\mu_1 + \frac{1}{2}\sigma_1^2(1 - \alpha)(2 - \alpha) + \lambda\zeta_1^{(\alpha-1)},$$

where the notation $\zeta_1^{(a)}$ means $\zeta_1^{(a)} := \mathbb{E}[(\tilde{V})^a - 1]$.

As will be seen in Proposition 1, this assumption basically guarantees that in equilibrium the risk-free interest rate is positive.

Proposition 1. Suppose $\zeta_1^{(\alpha-1)} < \infty$. (1) In this economy, the risk-free interest rate is constant. More precisely, letting $B(t, T)$ be the price of a zero coupon bond with maturity T , then $-\frac{1}{T-t}\log(B(t, T))$ is a constant independent of T , which will be called the risk-free interest rate r :

$$r = \theta + (1 - \alpha)\mu_1 - \frac{1}{2}\sigma_1^2(1 - \alpha)(2 - \alpha) - \lambda\zeta_1^{(\alpha-1)} > 0. \quad (11)$$

(2) Let $Z(t) := e^{rt}U_c(\delta(t), t) = e^{(r-\theta)t}(\delta(t))^{\alpha-1}$. Then $Z(t)$ is a martingale under $\mathbf{P}$,

$$\frac{dZ(t)}{Z(t-)} = -\lambda\zeta_1^{(\alpha-1)}dt + \sigma_1(\alpha - 1)dW_1(t) + d\left[\sum_{i=1}^{N(t)}(\tilde{V}_i^{\alpha-1} - 1)\right]. \quad (12)$$

Using $Z(t)$, one can define a probability measure $\mathbf{P}^*$: $\frac{d\mathbf{P}^*}{d\mathbf{P}} := Z(t)/Z(0)$. In terms of the notation $\mathbf{P}^*$, the Euler equation (10) holds if and only if

$$S(t) = e^{-r(T-t)}\mathbf{E}^*(S(T)|\mathcal{F}_t), \quad \forall T \geq t \geq 0. \quad (13)$$

Furthermore, the rational expectation equilibrium price of a European option (could be path-dependent), with the payoff $\psi_S(T)$ at the maturity T , is given by

$$\psi_S(t) = e^{-r(T-t)}\mathbf{E}^*(\psi_S(T)|\mathcal{F}_t), \quad \forall t \in [0, T]. \quad (14)$$

Proof. See Appendix A. $\square$

Given the endowment process $\delta(t)$, it must be decided what stochastic processes are suitable for $S(t)$ to satisfy the rational equilibrium requirement (10) or (13). I now postulate a special jump diffusion form for $S(t)$,

$$\frac{dS(t)}{S(t-)} = \mu dt + \sigma\{\rho dW_1(t) + \sqrt{1 - \rho^2}dW_2(t)\} + d\left(\sum_{i=1}^{N(t)}(V_i - 1)\right), \quad V_i = \tilde{V}_i^\beta, \quad (15)$$

where $W_2(t)$ is a Brownian motion independent of $W_1(t)$. In other words, the same Poisson process affects both the endowment $\delta(t)$ and $S(t)$, and the jump sizes are related through a

common power function, where the power $\beta \in (-\infty, \infty)$ is an arbitrary constant. The diffusion coefficients and the Brownian motion part for $\delta(t)$ and $S(t)$, though, are totally different.

It remains to see what the constraints on this particular model must be imposed, so that the jump diffusion model can be embedded in the rational expectation equilibrium requirement (10) or (13).

Theorem 1. Suppose $\zeta_1^{(\alpha+\beta-1)} < \infty$ and $\zeta_1^{(\alpha-1)} < \infty$. The model (15) satisfies the equilibrium requirement (13) if and only if

$$\begin{aligned}\mu &= r + \sigma_1 \sigma \rho (1 - \alpha) - \lambda (\zeta_1^{(\alpha+\beta-1)} - \zeta_1^{(\alpha-1)}) \\ &= \theta + (1 - \alpha) \left\{ \mu_1 - \frac{1}{2} \sigma_1^2 (2 - \alpha) + \sigma_1 \sigma \rho \right\} - \lambda \zeta_1^{(\alpha+\beta-1)}.\end{aligned}\quad (16)$$

If (16) is satisfied, then under P^*

$$\frac{dS(t)}{S(t-)} = r dt - \lambda^* E^*(\tilde{V}_i^\beta - 1) dt + \sigma dW^*(t) + d \left[\sum_{i=1}^{N(t)} (\tilde{V}_i^\beta - 1) \right]. \quad (17)$$

Here, under P^* , $W^*(t)$ is a new Brownian motion, $N(t)$ is a new Poisson process with jump rate $\lambda^* = \lambda E(\tilde{V}_i^{\alpha-1}) = \lambda (\zeta_1^{(\alpha-1)} + 1)$, and $\{\tilde{V}_i\}$ are independent identically distributed random variables with a new density under P^* :

$$f_{\tilde{V}}^*(x) = \frac{1}{\zeta_1^{(\alpha-1)} + 1} x^{\alpha-1} f_{\tilde{V}}(x). \quad (18)$$

Proof. See Appendix A. $\square$

The following corollary gives a condition under which all three dynamics, $\delta(t)$ and $S(t)$ under P , and $S(t)$ under P^* have the same jump diffusion form, which is very convenient for analytical calculation.

Corollary 1. If the family $\mathcal{V}$ of the distribution of the jump size $\tilde{V}$ for the endowment process $\delta(t)$ satisfies, for any real number a ,

$$\tilde{V}^a \in \mathcal{V}, \quad \text{const} \cdot x^a f_{\tilde{V}}(x) \in \mathcal{V}, \quad (19)$$

where the normalizing constant, const , is $\{\zeta_1^{(a-1)} + 1\}^{-1}$ (provided that $\zeta_1^{(a-1)} < \infty$), then the jump sizes for the asset price $S(t)$ under P and the jump sizes for $S(t)$ under the rational expectation risk-neutral measure P^* all belong to the same family $\mathcal{V}$.

Proof. Immediately follows from (9), (15), and (18). $\square$

The condition (19) essentially requires that the jump size distribution belongs to the exponential family. It is satisfied, for example, if $\log(V)$ has a normal distribution, or if $\log(V)$ has a double exponential distribution. However, the log power type distributions, such as log t -distribution, may not satisfy (19).

IV Option Pricing

In this section I will compute the rational expectation equilibrium option pricing formula (14) explicitly for European call and put options. For notation simplicity, I will drop $*$ in the risk-neutral notation, i.e. write η_1 instead of η_1^* , etc.

To compute (14), several results are needed. (1) One has to study the sum of the double exponential random variables; this is addressed in Appendix B.1. (2) One has to study the distribution of the sum of the double exponential random variables and normal random variables. Fortunately, this distribution can be obtained in closed form, thanks to a special function, the Hh function, from the mathematical physics literature.

IV.A Hh Functions

For every $n \geq 0$, the Hh function is a non-increasing function, defined as

$$\text{Hh}_n(x) = \int_x^\infty \text{Hh}_{n-1}(y) dy = \frac{1}{n!} \int_x^\infty (t-x)^n e^{-t^2/2} dt \geq 0, \quad n = 0, 1, 2, \dots \quad (20)$$

$$\text{Hh}_{-1}(x) = e^{-x^2/2} = \sqrt{2\pi}\varphi(x), \quad \text{Hh}_0(x) = \sqrt{2\pi}\Phi(-x);$$

see, for example, p. 691 of Abramowitz and Stegun (1972). The Hh function can be viewed as a generalization of the cumulative normal distribution⁸ function. Furthermore, the left tail of an Hh function has a polynomial growth rate, and the right tail has an exponential decay rate; see Appendix B.2.

The integral in (20) can be evaluated very fast⁹ by many software packages (for example, mathematica¹⁰). In addition,

$$\text{Hh}_n(x) = 2^{-n/2} \sqrt{\pi} e^{-x^2/2} \left\{ \frac{{}_1F_1(\frac{1}{2}n + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}x^2)}{\sqrt{2}\Gamma(1 + \frac{1}{2}n)} - x \frac{{}_1F_1(\frac{1}{2}n + 1, \frac{3}{2}, \frac{1}{2}x^2)}{\Gamma(\frac{1}{2} + \frac{1}{2}n)} \right\},$$

⁸For a standard normal random variable Z , $E((Z-x)^n I(Z \geq x)) = \frac{n!}{\sqrt{2\pi}} \text{Hh}_n(x)$, for $n \geq 0$.

⁹As a hypothetical example, it only takes a fractional CPU time on a Pentium 400 to get that $\text{Hh}_{100}[-1000] = 2.6992 * 10^{142}$.

¹⁰A short code (about 7 lines) can be downloaded from the author's web page. Note that the integrand in (20) has a maxima at $(x + \sqrt{x^2 + 4n})/2$; therefore, to numerically compute the integral, one should split the integral more around the maxima.

where ${}_1F_1$ is the confluent hypergeometric function; and a three term recursion¹¹ is also available for the Hh function (see pp. 299-300 and p. 691 of Abramowitz and Stegun, 1972):

$$nHh_n(x) = Hh_{n-2}(x) - xHh_{n-1}(x), \quad n \geq 1. \quad (21)$$

Therefore, one can compute all Hh_n functions, $n \geq 1$, effectively by using the normal density and normal distribution function. The following is a graph for the Hh functions.

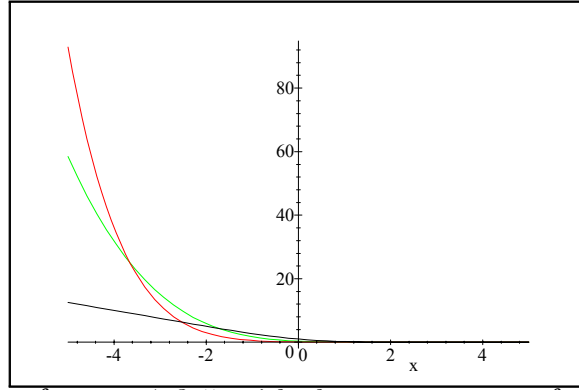


Figure 5: Hh functions for $n = 1, 3, 5$, with the steepest curve for $n = 3$ and the flattest curve for $n = 1$.

For option pricing it is important to evaluate the integral $I_n(\lambda; \alpha, \beta, \delta)$,

$$I_n(\lambda; \alpha, \beta, \delta) := \int_{\lambda}^{\infty} e^{\alpha x} Hh_n(\beta x - \delta) dx, \quad n \geq 0, \quad (22)$$

for arbitrary constants α , λ , and β .

Proposition 2. (1) If $\beta > 0$ and $\alpha \neq 0$, then for all $n \geq 0$,

$$I_n(\lambda; \alpha, \beta, \delta) = -\frac{e^{\alpha\lambda}}{\alpha} \sum_{i=0}^n \left(\frac{\beta}{\alpha}\right)^{n-i} Hh_i(\beta\lambda - \delta) + \left(\frac{\beta}{\alpha}\right)^{n+1} \frac{\sqrt{2\pi}}{\beta} e^{\frac{\alpha\delta}{\beta} + \frac{\alpha^2}{2\beta^2}} \Phi(-\beta\lambda + \delta + \frac{\alpha}{\beta}). \quad (23)$$

(2) If $\beta < 0$ and $\alpha < 0$, then for all $n \geq 0$,

$$I_n(\lambda; \alpha, \beta, \delta) = -\frac{e^{\alpha\lambda}}{\alpha} \sum_{i=0}^n \left(\frac{\beta}{\alpha}\right)^{n-i} Hh_i(\beta\lambda - \delta) - \left(\frac{\beta}{\alpha}\right)^{n+1} \frac{\sqrt{2\pi}}{\beta} e^{\frac{\alpha\delta}{\beta} + \frac{\alpha^2}{2\beta^2}} \Phi(\beta\lambda - \delta - \frac{\alpha}{\beta}). \quad (24)$$

Proof. See¹² the Appendix B.2. $\square$

¹¹The solution of this recursion, though lengthy, can be written in terms of the absolute value of the coefficients in Hermit polynomials (whence the name Hh).

¹²If $\beta > 0$ and $\alpha = 0$ then for all $n \geq 0$, $I_n(\lambda; \alpha, \beta, \delta) = \frac{1}{\beta} Hh_{n+1}(\beta\lambda - \delta)$. If $\beta \leq 0$ and $\alpha \geq 0$, then for all $n \geq 0$, $I_n(\lambda; \alpha, \beta, \delta) = \infty$. If $\beta = 0$ and $\alpha < 0$, then for all $n \geq 0$, $I_n(\lambda; \alpha, \beta, \delta) = \int_{\lambda}^{\infty} e^{\alpha x} Hh_n(-\delta) dx = Hh_n(-\delta)e^{\alpha\lambda}$.

IV.B European Call and Put Options

Theorem 2. The rational equilibrium price of a European call option from (14) is given by

$$\begin{aligned}
\psi_c(0) = & S(0) a_1 \sum_{n=1}^{\infty} \pi_n \sum_{k=1}^n P_{n,k}(\sigma\sqrt{T}\eta_1)^k I_{k-1}(h; 1 - \eta_1, -\frac{1}{\sigma\sqrt{T}}, -\sigma\eta_1\sqrt{T}) \\
& - K e^{-rT} d_1 \sum_{n=1}^{\infty} \pi_n \sum_{k=1}^n P_{n,k}(\sigma\sqrt{T}\eta_1)^k I_{k-1}(h; -\eta_1, -\frac{1}{\sigma\sqrt{T}}, -\sigma\eta_1\sqrt{T}) \\
& + S(0) a_2 \sum_{n=1}^{\infty} \pi_n \sum_{k=1}^n Q_{n,k}(\sigma\sqrt{T}\eta_2)^k I_{k-1}(h; 1 + \eta_2, \frac{1}{\sigma\sqrt{T}}, -\sigma\eta_2\sqrt{T}) \\
& - K e^{-rT} d_2 \sum_{n=1}^{\infty} \pi_n \sum_{k=1}^n Q_{n,k}(\sigma\sqrt{T}\eta_2)^k I_{k-1}(h; \eta_2, \frac{1}{\sigma\sqrt{T}}, -\sigma\eta_2\sqrt{T}) \\
& + \pi_0 \{S(0)e^{-\lambda\zeta T} \Phi(b_+) - K e^{-rT} \Phi(b_-)\}.
\end{aligned}$$

where

$$\begin{aligned}
b_{\pm} &= \frac{\log(S(0)/K) + \left(r \pm \frac{\sigma^2}{2} - \lambda\zeta\right) T}{\sigma\sqrt{T}}, \quad \zeta = \frac{p\eta_1}{\eta_1 - 1} + \frac{q\eta_2}{\eta_2 + 1} - 1, \\
a_i &= e^{-(\lambda\zeta + \frac{\sigma^2}{2})T} d_i, \quad d_i = \frac{e^{(\sigma\eta_i)^2 T/2}}{\sigma\sqrt{2\pi T}}, \quad i = 1, 2, \quad h = \log\left(\frac{K}{S(0)}\right) + \lambda\zeta T - \left(r - \frac{\sigma^2}{2}\right) T, \\
P_{n,k} &= \sum_{i=k}^{n-1} \binom{n-k-1}{i-k} \binom{n}{i} \cdot \left(\frac{\eta_1}{\eta_1 + \eta_2}\right)^{i-k} \left(\frac{\eta_2}{\eta_1 + \eta_2}\right)^{n-i} p^i q^{n-i}, \quad 1 \leq k \leq n-1, \\
Q_{n,k} &= \sum_{i=k}^{n-1} \binom{n-k-1}{i-k} \binom{n}{i} \cdot \left(\frac{\eta_1}{\eta_1 + \eta_2}\right)^{n-i} \left(\frac{\eta_2}{\eta_1 + \eta_2}\right)^{i-k} p^{n-i} q^i, \quad 1 \leq k \leq n-1, \\
P_{n,n} &= p^n, \quad Q_{n,n} = q^n, \quad \pi_n := P^*(N(T) = n) = e^{-\lambda T} (\lambda T)^n / n!,
\end{aligned}$$

and $\binom{0}{0}$ is defined to be 1. The price of a put option, $\psi_p(0)$, can be obtained by the put-call parity:

$$\psi_p(0) - \psi_c(0) = e^{-rT} \mathbf{E}^*((K - S(T))^+ - (S(T) - K)^+) = e^{-rT} \mathbf{E}^*(K - S(T)) = K e^{-rT} - S(0).$$

Proof. See¹³ the Appendices B.3 and B.4. $\square$

¹³Although the pricing formulae involve infinite series, numerically the expressions can be evaluated very quickly to a high degree of accuracy through truncation. My experience suggests that numerically only the first 10 to 15 terms in the infinite series are needed for most applications.

Now I consider the problem of pricing options on futures contracts. Assume, for now, the term structure of interest rate is flat, and r is a constant. Then the futures price, $F(t, T^*)$, with maturity at time T^* , is given by $F(t, T^*) = E^*(S(T^*)|\mathcal{F}_t) = e^{r(T^*-t)}S(t)$.

Corollary 2. The price of a European call option, $\psi_{c,F}(D, F(0, T^*), T)$, on a futures contract is given by

$$\begin{aligned} \psi_{c,F}(D, F(0, T^*), T) = & D \cdot [F(0, T^*)a_1 \sum_{n=1}^{\infty} \pi_n \sum_{k=1}^n P_{n,k}(\sigma\sqrt{T}\eta_1)^k I_{k-1}(h_F; 1 - \eta_1, -\frac{1}{\sigma\sqrt{T}}, -\sigma\eta_1\sqrt{T}) \\ & - Kd_1 \sum_{n=1}^{\infty} \pi_n \sum_{k=1}^n P_{n,k}(\sigma\sqrt{T}\eta_1)^k I_{k-1}(h_F; -\eta_1, -\frac{1}{\sigma\sqrt{T}}, -\sigma\eta_1\sqrt{T}) \\ & + F(0, T^*)a_2 \sum_{n=1}^{\infty} \pi_n \sum_{k=1}^n Q_{n,k}(\sigma\sqrt{T}\eta_2)^k I_{k-1}(h_F; 1 + \eta_2, \frac{1}{\sigma\sqrt{T}}, -\sigma\eta_2\sqrt{T}) \\ & - Kd_2 \sum_{n=1}^{\infty} \pi_n \sum_{k=1}^n Q_{n,k}(\sigma\sqrt{T}\eta_2)^k I_{k-1}(h_F; \eta_2, \frac{1}{\sigma\sqrt{T}}, -\sigma\eta_2\sqrt{T}) \\ & + \pi_0\{F(0, T^*)e^{-\lambda\zeta T}\Phi(b_{+,F}) - K\Phi(b_{-,F})\}], \end{aligned}$$

where

$$b_{\pm,F} = \frac{\log(F(0, T^*)/K) \pm \frac{\sigma^2}{2}T - \lambda\zeta T}{\sigma\sqrt{T}}, \quad h_F = \log(K/F(0, T^*)) + \lambda\zeta T + \frac{\sigma^2}{2}T, \quad D = e^{-rT}.$$

The put option can be priced according to the put-call parity:

$$\psi_{p,F}(D, F(0, T^*), T) - \psi_{c,F}(D, F(0, T^*), T) = e^{-rT}(K - F(0, T^*)).$$

This corollary can be proved easily by using Theorem 2 and standard proofs about the futures options. Hence the proof is omitted. Note that, however, Corollary 2 cannot simply be proved by plunging in the futures formula into Theorem 2, as the underlying asset now is the futures contract rather than the spot asset itself.

Corollary 3. (1) If the jump size getting smaller and smaller, then the pricing formulae in Theorem 2 and Corollary 2 degenerate to the Black-Scholes formula options and Black's futures formula. More precisely, as both $\eta_1 \rightarrow \infty$ and $\eta_2 \rightarrow \infty$ while all other parameters being fixed,

$$\psi_c(0) \rightarrow S(0)\Phi(b'_+) - Ke^{-rT}\Phi(b'_-), \quad \psi_{c,F}(0) \rightarrow e^{-rT}\{F(0, T^*)\Phi(b'_{+,F}) - K\Phi(b'_{-,F})\},$$

where

$$b'_\pm := \frac{\log(S(0)/K) + \left(r \pm \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}, \quad b'_{\pm,F} = \frac{\log(F(0,T^*)/K) \pm \sigma^2 T/2}{\sigma\sqrt{T}}.$$

(2) If the jump rate is zero, i.e. $\lambda = 0$, then the pricing formulae again degenerate to the Black-Scholes and Black's futures formulae.

Proof. See Appendix B.4. $\square$

V Pricing of Interest Rate Derivatives and “Volatility Smile”

V.A Pricing Interest Rate Caps and Floors

Interest rates with jumps have been studied by many papers¹⁴. In particular, a general framework of pricing interest rate swaptions, caps and floors with jump risk has been studied in detail in Glasserman and Kou (1999). The results in this section is a corollary of that paper, with the logarithm of the jump sizes being specialized to be double exponentially distributed, except that one has to do the explicit calculation in the case of the double exponential distribution. Therefore, I will only give a sketch of the results of caps and floors, and refer the readers to Section 7 in Glasserman and Kou (1999) for pricing of swaptions.

Interest rate caps and floors are among the most liquid of all interest rate derivatives. To study them, it is necessary to consider the term structures based on simple forward rates. With a fixed accrual period δ expressed as a fraction of a year (e.g., $\delta = 1/4$ means three month rates), let $L(t, T)$ be the simple forward rate for the interval from T to $T + \delta$ as of time $t \leq T$. Mathematically speaking, the forward rate is defined as $L(t, T) = \frac{1}{\delta} \left(\frac{B(t, T)}{B(t, T+\delta)} - 1 \right)$, where $B(t, \tau)$ is the time- t price of a zero coupon bond maturing at time τ . An interest rate caplet for the period $[T_n, T_{n+1}]$, with $T_n = n\delta$, and struck at K is a derivative security paying $\delta(L_n(T_n) - K)^+$ at time T_{n+1} , where $L_n(t) = L(t, n\delta)$.

Under the marked-point-processes model in Glasserman and Kou (1999), particularly equation (9) in that paper with $\log X_n^{(i)}$ being double exponential random variables instead of being normal, the time- t price $C_n(t)$ of the n th caplet can be obtained explicitly, if (23) in Glasserman and Kou (1999) holds with f_n a double exponential density. In particular, if σ_k are deterministic

¹⁴Equilibrium asset pricing models for interest rates with jumps include Ahn and Thompson (1988), Das and Foresi (1996); the pricing of interest rate derivatives in the presence of jumps is considered in Björk, Kabanov, and Runggaldier (1997), Das and Foresi (1996), Duffie, Pan, and Singleton (2000), Jarrow and Madan (1995), and Shirakawa (1991). Various sources of jumps in interest rates, including moves by central banks, are investigated in Babbs and Webber (1997), El-Jahel, Lindberg, and Perraudin (1997).

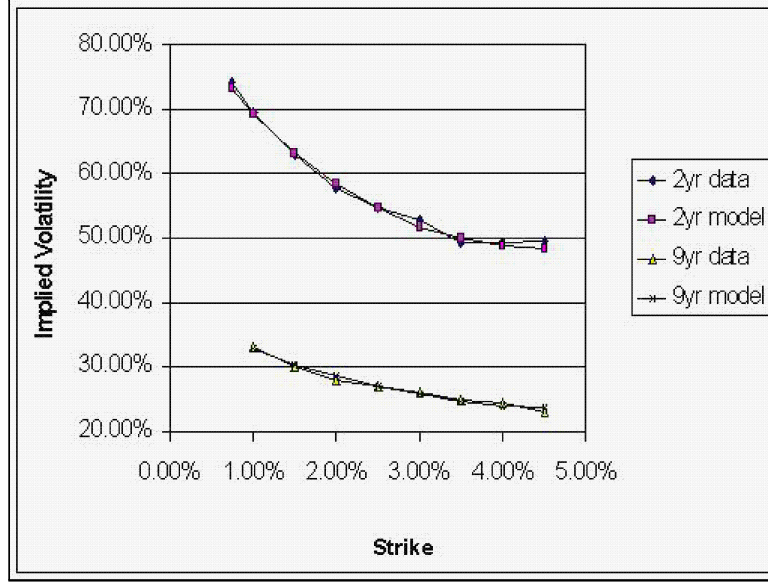


Figure 6: Mid-market and model implied volatilities for Japanese LIBOR caplets in May 1998. The parameters used for the model implied volatilities are $\eta_4^{(1)} = 3.7$, $\eta_4^{(2)} = 1.8$, $p_4 = 0.04$, $\lambda_4 = 1.4$, $\sigma_4 = 0.21$, and $\eta_{18}^{(1)} = 2.3$, $\eta_{18}^{(2)} = 1.8$, $p_{18} = 0.09$, $\lambda_{18} = 0.2$, $\sigma_{18} = 0.09$.

then the time- t price of the n th caplet, $t < T_n$, is given by

$$C_n(t) = \delta \psi_{c,F}(B_{n+1}(t), L_n(t), T_n - t), \quad (25)$$

in the notation of Corollary 2. Obviously, for $T_n \leq t \leq T_{n+1}$, $C_n(t) = \delta B_{n+1}(t)(L_n(T_n) - K)^+$. In other words, one can price interest rate caplets by using the formulae for call options on the futures contracts. By summing the prices of individual caplets one can price a cap, which is simply a portfolio of caplets with consecutive maturities. Similarly, one can price an interest rate floor by summing the prices of single-period floors, via the formulae for put options on futures contracts.

As noted in, for example, Miltersen, Sandmann, and Sondermann (1997), from the prices of caps and floors it is possible to derive prices of puts and calls on zero coupon bonds, provided the maturity of the bond is one period later in the tenor structure than the expiration of the option. In particular, a put option on the bond struck at K can be valued as a portfolio of K caplets struck at $(1 - K)/(\delta K)$.

V.B Implied “Volatility Smile”

To illustrate that the model can produce “implied volatility smile”, I consider a real data set used first in Andersen and Andreasen (2000) for 2-year and 9-year caplets in the Japanese LIBOR market as of late May 1998. In this example, I choose a set of model parameters, calculate caplet prices at a range of strikes using (25), and then find the corresponding implied volatilities based on the Black’s formula. Their implied volatilities based on mid-market prices are reproduced in Figure 6. Corresponding 6-month forward rates for the same period are 1.181% for the 2-year maturity and 2.913% for the 9-year maturity.¹⁵ Figure 6 also shows implied volatility curves derived from (25) using the same forward rates and the parameters in the caption. (With 6-month accrual intervals, the 2-year and 9-year caplets correspond to $n = 4$ and $n = 18$ in the notation of (25).)

Because the parameters used in (25) apply under a risk-neutral (rational expectation equilibrium) measure rather than the physical measure, we can interpret the parameters in Figure 6 as suggesting that the market attaches a large risk premium to the possibility of a sharp downward movement in Japanese interest rates. A similar interpretation is also given in Glasserman and Kou (1999) for the normal jump diffusion model.

I should emphasize that this example is only meant to be a test of the model; it is only served as an illustration to show that the model can produce a close fit even to a very sharp volatility skew.

VI Conclusion and Discussion

The paper proposes a double exponential jump diffusion model, in which the price of the underlying asset is modeled by two parts, a continuous part driven by Brownian motion, and a jump part with the logarithm of jump sizes having a double exponential distribution. The major advantages of the model are: (a) It can lead to the asymmetric leptokurtic feature and to the implied “volatility smile”. (b) It has a good analytical tractability; namely it can lead to analytic solutions for various option pricing problems, including path-dependent options and interest rate derivatives; in addition, the parameters in the model can be easily interpreted. (c) It can be embedded in a rational expectation equilibrium framework, thus self-consistent. (d) It has a psychological interpretation.

¹⁵I thank Leif Andersen of General Re Financial Products for providing these values and also the mid-market implied volatilities in the figure.

The closed form solutions are made possible because of the simplicity of the exponential distribution and the memoryless property. In Kou and Wang (2000, 2001), analytical solutions for various path-dependent options, such as perpetual American options, barrier and lookback options are presented. These options cannot be priced in closed form if the logarithm of the jump sizes have a normal distribution, because of the overshoot problem related to the boundary crossing problems, as mentioned in Section I.B.

One disadvantage of the model is that the pricing formulae, though being analytic, appear to be long. However, this may not be a major problem because the Hh function can be computed easily via a linear recursion (21), and what appears to be lengthy to human eyes might make little difference in terms of computer programming, as long as it is a closed form solution.

A serious criticism is perhaps about hedging. Due to the jump part, the market is incomplete, and the conventional riskless hedging arguments are not applicable here. However, it should be pointed out that the riskless hedging is really a special property of continuous time Brownian motion, and it does not hold for most of the alternative models. Even within the Brownian motion framework, the riskless hedging is impossible if one wants to do it only in discrete time. For some suggestions about hedging with jump risk, see, for example, Merton (1976) and Naik and Lee (1990).

I should also mention that no empirical test has been conducted in this paper. However, the empirical tests performed in Ramezani and Zeng (1998) suggest that the double exponential jump diffusion model fits stock data better than the normal jump diffusion model. If one wants to see how much empirical improvement that can be made by including both jumps and stochastic volatility, the testing method proposed in Andersen, Benzoni, and Lund (1999) will be useful.

A Appendix: Derivation of the Rational Expectation

Proof of Proposition 1. (1) Since $B(T, T) = 1$, equation (10) yields

$$B(t, T) = e^{-\theta(T-t)} \frac{\mathbb{E}((\delta(T))^{\alpha-1} | \mathcal{F}_t)}{(\delta(t))^{\alpha-1}}. \quad (26)$$

Using the facts that

$$\left(\frac{\delta(T)}{\delta(t)} \right)^{\alpha-1} = \exp\left\{(\alpha-1)\left(\mu_1 - \frac{1}{2}\sigma_1^2\right)(T-t) + \sigma_1(\alpha-1)(W_1(T) - W_1(t))\right\} \prod_{i=N(t)+1}^{N(T)} \tilde{V}_i^{\alpha-1},$$

$$\mathbb{E}\left(\prod_{i=N(t)+1}^{N(T)} \tilde{V}_i^{\alpha-1}\right) = \sum_{j=0}^{\infty} e^{-\lambda(T-t)} \frac{[\lambda(T-t)]^j}{j!} \{\zeta_1^{(\alpha-1)} + 1\}^j = \exp\{\lambda\zeta_1^{(\alpha-1)}(T-t)\},$$

equation (26) yields

$$B(t, T) = \exp[-(T-t)\{\theta - (\alpha-1)(\mu_1 - \frac{1}{2}\sigma_1^2) - \frac{1}{2}\sigma_1^2(\alpha-1)^2 - \lambda\zeta_1^{(\alpha-1)}\}],$$

from which (11) follows.

(2) Note that (26) implies

$$e^{-r(T-t)} = \mathbb{E}(U_c(\delta(T), T)/U_c(\delta(t), t)|\mathcal{F}_t), \quad (27)$$

which shows that $Z(t)$ is a martingale under the probability $\mathbb{P}$. Furthermore, (9) and (11) lead to

$$\begin{aligned} Z(t) &= e^{(r-\theta)t} (\delta(t))^{\alpha-1} \\ &= (\delta(0))^{\alpha-1} e^{(r-\theta)t} \exp\{(\alpha-1)(\mu_1 - \frac{1}{2}\sigma_1^2)t + \sigma_1(\alpha-1)W_1(t)\} \prod_{i=1}^{N(t)} \tilde{V}_i^{\alpha-1} \\ &= (\delta(0))^{\alpha-1} \exp\{-\frac{1}{2}\sigma_1^2(\alpha-1)^2 - \lambda\zeta_1^{(\alpha-1)}\}t + \sigma_1(\alpha-1)W_1(t)\} \prod_{i=1}^{N(t)} \tilde{V}_i^{\alpha-1}, \end{aligned}$$

from which (12) follows. Now by (10) and the interest rate formula (27),

$$\begin{aligned} S(t) &= \frac{\mathbb{E}(U_c(\delta(T), T)S(T)|\mathcal{F}_t)}{U_c(\delta(t), t)} = e^{-r(T-t)} \mathbb{E}\left\{\frac{Z(T)}{Z(t)} S(T) | \mathcal{F}_t\right\} = e^{-r(T-t)} \mathbb{E}^*(S(T)|\mathcal{F}_t), \\ \psi_S(t) &= \frac{\mathbb{E}(U_c(\delta(T), T)\psi_S(T)|\mathcal{F}_t)}{U_c(\delta(t), t)} = e^{-rT} \mathbb{E}\left\{\frac{Z(T)}{Z(t)} \psi_S(T) | \mathcal{F}_t\right\} = e^{-rT} \mathbb{E}^*(\psi_S(T)|\mathcal{F}_t). \end{aligned}$$

Proof of Theorem 1. Girsanov theorem for jump diffusion processes tells us that under $\mathbb{P}^*$, $W'_1(t) := W_1(t) - \sigma_1(\alpha-1)t$ is a new Brownian motion, and under $\mathbb{P}^*$ the jump rate of $N(t)$ is $\lambda^* = \lambda\mathbb{E}(\tilde{V}_i^{\alpha-1}) = \lambda(\zeta_1^{(\alpha-1)} + 1)$, and $\tilde{V}_i$ has a new density $f_{\tilde{V}}^*(x) = \frac{1}{\zeta_1^{(\alpha-1)} + 1} x^{\alpha-1} f_{\tilde{V}}(x)$. Therefore, the dynamics of $S(t)$ is given by

$$\begin{aligned} \frac{dS(t)}{S(t-)} &= \mu dt + \sigma\{\rho dW_1(t) + \sqrt{1-\rho^2}dW_2(t)\} + \Delta \left[\sum_{i=1}^{N(t)} (\tilde{V}_i^\beta - 1) \right] \\ &= \{\mu + \sigma_1\sigma\rho(\alpha-1)\}dt + \sigma\{\rho dW'_1(t) + \sqrt{1-\rho^2}dW_2(t)\} + \Delta \left[\sum_{i=1}^{N(t)} (\tilde{V}_i^\beta - 1) \right]. \end{aligned}$$

Since

$$\mathbb{E}^*(\tilde{V}_i^\beta) = \int_0^\infty x^\beta \frac{1}{\zeta_1^{(\alpha-1)} + 1} x^{\alpha-1} f_{\tilde{V}}(x) dx = \frac{1}{\zeta_1^{(\alpha-1)} + 1} \mathbb{E}(\tilde{V}^{\alpha+\beta-1}) = \frac{\zeta_1^{(\alpha+\beta-1)} + 1}{\zeta_1^{(\alpha-1)} + 1},$$

we have

$$\lambda^* \{E^*(\tilde{V}_i^\beta) - 1\} = \lambda^* \left\{ \frac{\zeta_1^{(\alpha+\beta-1)} + 1}{\zeta_1^{(\alpha-1)} + 1} - 1 \right\} = \lambda(\zeta_1^{(\alpha+\beta-1)} - \zeta_1^{(\alpha-1)}).$$

Therefore,

$$\begin{aligned} \frac{dS(t)}{S(t-)} &= \{\mu + \sigma_1 \sigma \rho(\alpha - 1) + \lambda(\zeta_{\alpha+\beta-1} - \zeta_{\alpha-1})\} dt \\ &\quad - \lambda^* \{E^*(\tilde{V}_i^\beta) - 1\} dt + \sigma \{\rho dW_1'(t) + \sqrt{1 - \rho^2} dW_2(t)\} + \Delta \left[\sum_{i=1}^{N(t)} (\tilde{V}_i^\beta - 1) \right]. \end{aligned}$$

Hence, to satisfy the rational equilibrium requirement $S(t) = e^{-r(T-t)} E^*(S(T) | \mathcal{F}_t)$, we must have $\mu + \sigma_1 \sigma \rho(\alpha - 1) + \lambda(\zeta_{\alpha+\beta-1} - \zeta_{\alpha-1}) = r$, from which (16) follows. If (16) is satisfied, under the measure P^* , the dynamics for $S(t)$ is given by

$$\frac{dS(t)}{S(t-)} = r dt - \lambda^* \{E^*(\tilde{V}_i^\beta) - 1\} dt + \sigma \{\rho dW_1'(t) + \sqrt{1 - \rho^2} dW_2(t)\} + \Delta \left[\sum_{i=1}^{N(t)} (\tilde{V}_i^\beta - 1) \right],$$

from which (17) follows. $\square$

B Appendix: Derivation of the Option Pricing Formulae

B.1 Decomposition of the Sum of Double Exponential Random Variables

The memoryless property of exponential random variables yields

$$(\xi^+ - \xi^- | \xi^+ > \xi^-) \stackrel{d}{=} \xi^+, \quad (\xi^+ - \xi^- | \xi^+ < \xi^-) \stackrel{d}{=} -\xi^-,$$

thus leading to the conclusion that

$$\xi^+ - \xi^- \stackrel{d}{=} \begin{cases} \xi^+, & \text{with probability } \eta_2/(\eta_1 + \eta_2) \\ -\xi^-, & \text{with probability } \eta_1/(\eta_1 + \eta_2) \end{cases}, \quad (28)$$

because the probabilities of the events $\xi^+ > \xi^-$ and $\xi^+ < \xi^-$ are $\eta_2/(\eta_1 + \eta_2)$ and $\eta_1/(\eta_1 + \eta_2)$, respectively. The following proposition extends (28).

Proposition B.1. For every $n \geq 1$, we have the following decomposition

$$\sum_{i=1}^n Y_i \stackrel{d}{=} \begin{cases} \sum_{i=1}^k \xi_i^+, & \text{with probability } P_{n,k}, \quad k = 1, 2, \dots, n \\ -\sum_{i=1}^k \xi_i^-, & \text{with probability } Q_{n,k}, \quad k = 1, 2, \dots, n \end{cases}, \quad (29)$$

where $P_{n,k}$ and $Q_{n,k}$ are defined in Theorem 2. Here ξ_i^+ and ξ_i^- are i.i.d. exponential random variables with rates η_1 and η_2 , respectively.

As a key step in deriving closed form solutions for call and put options, this proposition indicates that the sum of i.i.d. double exponential random variables can be written, in distribution, as a (randomly) mixed gamma random variable¹⁶. To prove Proposition B.1, the following lemma is needed.

Lemma B.1.

$$\sum_{i=1}^n \xi_i^+ - \sum_{j=1}^m \xi_j^- \stackrel{d}{=} \left\{ \begin{array}{ll} \sum_{i=1}^k \xi_i, & \text{with prob. } \left(\frac{\eta_1}{\eta_1+\eta_2}\right)^{n-k} \left(\frac{\eta_2}{\eta_1+\eta_2}\right)^m \cdot \binom{n-k+m-1}{m-1}, k = 1, \dots, n \\ -\sum_{i=1}^l \xi_i, & \text{with prob. } \left(\frac{\eta_1}{\eta_1+\eta_2}\right)^n \left(\frac{\eta_2}{\eta_1+\eta_2}\right)^{m-l} \cdot \binom{n-l+m-1}{n-1}, l = 1, \dots, m \end{array} \right\}. \quad (30)$$

Proof. Introduce the random variables $A(n, m) = \sum_{i=1}^n \xi_i - \sum_{j=1}^m \tilde{\xi}_j$. Then

$$A(n, m) \stackrel{d}{=} \left\{ \begin{array}{ll} A(n-1, m-1) + \xi^+, & \eta_2/(\eta_1 + \eta_2) \\ A(n-1, m-1) - \xi^-, & \eta_1/(\eta_1 + \eta_2) \end{array} \right\} \stackrel{d}{=} \left\{ \begin{array}{ll} A(n, m-1), & \eta_2/(\eta_1 + \eta_2) \\ A(n-1, m), & \eta_1/(\eta_1 + \eta_2) \end{array} \right\},$$

via (28). Now image a plane with the horizontal axis representing the number of $\{\xi_i^+\}$ and vertical axis representing the number of $\{\xi_j^-\}$. Suppose we have a random walk on the integer lattice point of this plane. Starting from any point (n, m) , $n, m \geq 1$, the random walk goes either one step to the left with probability $\eta_1/(\eta_1 + \eta_2)$ or one step down with equal probability $\eta_2/(\eta_1 + \eta_2)$, and the random walk stops once it reaches either the horizontal or the vertical axis. For any path that connecting (n, m) to $(k, 0)$, $1 \leq k \leq n$, it must reach $(k, 1)$ first before it makes a final step to $(k, 0)$. Furthermore, all the paths going from (n, m) to $(k, 1)$ must have exactly $n - k$ left's and $m - 1$ down's, whence the total number of such paths is $\binom{n-k+m-1}{m-1}$. Similarly, the total number of paths that leading point (n, m) to $(0, l)$, $1 \leq l \leq m$, is $\binom{n-l+m-1}{n-1}$. Thus,

$$A(n, m) \stackrel{d}{=} \left\{ \begin{array}{ll} \sum_{i=1}^k \xi_i, & \text{with prob. } \left(\frac{\eta_1}{\eta_1+\eta_2}\right)^{n-k} \left(\frac{\eta_2}{\eta_1+\eta_2}\right)^m \cdot \binom{n-k+m-1}{m-1}, k = 1, \dots, n \\ -\sum_{i=1}^l \xi_i, & \text{with prob. } \left(\frac{\eta_1}{\eta_1+\eta_2}\right)^n \left(\frac{\eta_2}{\eta_1+\eta_2}\right)^{m-l} \cdot \binom{n-l+m-1}{n-1}, l = 1, \dots, m \end{array} \right\},$$

and the lemma is proven. $\square$

Proof of Proposition B.1. By the same analogy used in Lemma 1, to compute probability $P_{n,k}$, $1 \leq k \leq n$, the probability weight assigned to $\sum_{i=1}^k \xi_i^+$ when we decompose $\sum_{i=1}^n Y_i$, it is equivalent to consider the probability of the random walk ever reach $(k, 0)$ starting from the point $(i, n-i)$, $0 \leq i \leq n$, with probability of starting from $(i, n-i)$ being $\binom{n}{i} p^i q^{n-i}$. Note that the point $(k, 0)$ can only be reached from points $(i, n-i)$, with the constraint $k \leq i \leq n-1$,

¹⁶A general result similar to the decomposition (29) was first discovered in Shanthikumar (1985), although (29) gives a more explicit calculation of $P_{n,k}$ and $Q_{n,k}$. Furthermore, the proofs are totally different: a combinatorial approach is used here, while the proof in Shanthikumar (1985) is based on the Laplace transform.

because the random walk can only go left or down, and stops once it reaches the horizontal axis. Therefore, for $1 \leq k \leq n-1$, (30) leads to

$$\begin{aligned}
P_{n,k} &= \sum_{i=k}^{n-1} \mathbf{P}(\text{going from } (i, n-i) \text{ to } (k, 0)) \cdot \mathbf{P}(\text{starting from } (i, n-i)) \\
&= \sum_{i=k}^{n-1} \left(\frac{\eta_1}{\eta_1 + \eta_2} \right)^{i-k} \left(\frac{\eta_2}{\eta_1 + \eta_2} \right)^{n-i} \binom{i + (n-i) - k - 1}{(n-i) - 1} \cdot \binom{n}{i} p^i q^{n-i} \\
&= \sum_{i=k}^{n-1} \binom{n-k-1}{n-i-1} \binom{n}{i} \left(\frac{\eta_1}{\eta_1 + \eta_2} \right)^{i-k} \left(\frac{\eta_2}{\eta_1 + \eta_2} \right)^{n-i} p^i q^{n-i} \\
&= \sum_{i=k}^{n-1} \binom{n-k-1}{i-k} \binom{n}{i} \left(\frac{\eta_1}{\eta_1 + \eta_2} \right)^{i-k} \left(\frac{\eta_2}{\eta_1 + \eta_2} \right)^{n-i} p^i q^{n-i},
\end{aligned}$$

where $\binom{0}{0}$ is defined to be one. Of course, $P_{n,n} = p^n$. Similarly, to compute $Q_{n,k}$, $k \geq 1$, which is the probability weight assigned to $-\sum_{i=1}^k \xi_i^-$ when we decompose $\sum_{i=1}^n Y_i$, it is equivalent to consider the probability of the random walk ever reach $(0, k)$ from the point $(n-i, i)$, $0 \leq i \leq n$. This is given by

$$\begin{aligned}
Q_{n,k} &= \sum_{i=k}^{n-1} \mathbf{P}(\text{going from } (n-i, i) \text{ to } (0, k)) \cdot \mathbf{P}(\text{starting from } (n-i, i)) \\
&= \sum_{i=k}^{n-1} \left(\frac{\eta_1}{\eta_1 + \eta_2} \right)^{n-i} \left(\frac{\eta_2}{\eta_1 + \eta_2} \right)^{i-k} \binom{(n-i) + i - k - 1}{(n-i) - 1} \cdot \binom{n}{n-i} p^{n-i} q^i \\
&= \sum_{i=k}^{n-1} \binom{n-k-1}{i-k} \binom{n}{i} \left(\frac{\eta_1}{\eta_1 + \eta_2} \right)^{n-i} \left(\frac{\eta_2}{\eta_1 + \eta_2} \right)^{i-k} p^{n-i} q^i,
\end{aligned}$$

with $Q_{n,n} = q^n$. Incidentally, we have also shown that $\sum_{k=1}^n (P_{n,k} + Q_{n,k}) = 1$. $\square$

B.2 Results on Hh Functions

First of all, note that $Hh_n(x) \rightarrow 0$, as $x \rightarrow \infty$, for $n \geq -1$; and $Hh_n(x) \rightarrow \infty$, as $x \rightarrow -\infty$, for $n \geq 1$; and $Hh_0(x) = \sqrt{2\pi}\Phi(-x) \rightarrow \sqrt{2\pi}$, as $x \rightarrow -\infty$. Note that for every $n \geq -1$, as $x \rightarrow \infty$,

$$\lim_{x \rightarrow \infty} Hh_n(x) / \left\{ \frac{1}{x^{n+1}} e^{-x^2/2} \right\} = 1. \quad (31)$$

and as $x \rightarrow -\infty$,

$$Hh_n(x) = O(|x|^n). \quad (32)$$

Here (31) follows from (19.14.3 and 19.8.1 in Abramowitz and Stegun, 1972); and (32) is clearly true for $n = -1$, and for $n \geq 0$ note that as $x \rightarrow -\infty$,

$$Hh_n(x) = \frac{1}{n!} \int_x^\infty (t-x)^n e^{-t^2/2} dt \leq \frac{2^n}{n!} \int_{-\infty}^\infty |t|^n e^{-t^2/2} dt + \frac{2^n}{n!} \int_{-\infty}^\infty |x|^n e^{-t^2/2} dt = O(|x|^n).$$

Proof of Proposition 2. Case 1: when $\beta > 0$ and $\alpha \neq 0$. In this case, we must have, for $n \geq 0$,

$$e^{\alpha x} \text{Hh}_n(\beta x - \delta) \rightarrow 0, \text{ as } x \rightarrow \infty, \quad (33)$$

for any constant α , thanks to (31). Integration by parts leads to

$$I_n = \frac{1}{\alpha} \int_{\lambda}^{\infty} \text{Hh}_n(\beta x - \delta) de^{\alpha x} = -\frac{1}{\alpha} \text{Hh}_n(\beta \lambda - \delta) e^{\alpha \lambda} + \frac{\beta}{\alpha} \int_{\lambda}^{\infty} e^{\alpha x} \text{Hh}_{n-1}(\beta x - \delta) dx,$$

where the limiting behavior in (33) has been applied. In other words, we have a recursion, for $n \geq 0$,

$$I_n = -\frac{e^{\alpha \lambda}}{\alpha} \text{Hh}_n(\beta \lambda - \delta) + \frac{\beta}{\alpha} I_{n-1},$$

with the initial condition

$$I_{-1} = \sqrt{2\pi} \int_{\lambda}^{\infty} e^{\alpha x} \varphi(-\beta x + \delta) dx = \frac{\sqrt{2\pi}}{\beta} \exp\left\{\frac{\alpha \delta}{\beta} + \frac{\alpha^2}{2\beta^2}\right\} \Phi\left(-\beta \lambda + \delta + \frac{\alpha}{\beta}\right).$$

Solving it yields, for $n \geq -1$,

$$\begin{aligned} I_n &= -\frac{e^{\alpha \lambda}}{\alpha} \sum_{i=0}^n \left(\frac{\beta}{\alpha}\right)^i \text{Hh}_{n-i}(\beta \lambda - \delta) + \left(\frac{\beta}{\alpha}\right)^{n+1} I_{-1} \\ &= -\frac{e^{\alpha \lambda}}{\alpha} \sum_{i=0}^n \left(\frac{\beta}{\alpha}\right)^{n-i} \text{Hh}_i(\beta \lambda - \delta) + \left(\frac{\beta}{\alpha}\right)^{n+1} \frac{\sqrt{2\pi}}{\beta} \exp\left\{\frac{\alpha \delta}{\beta} + \frac{\alpha^2}{2\beta^2}\right\} \Phi\left(-\beta \lambda + \delta + \frac{\alpha}{\beta}\right), \end{aligned}$$

where the sum over an empty set is defined to be zero.

Case 2: when $\beta < 0$ and $\alpha < 0$. In this case, we must also have, for $n \geq 0$,

$$e^{\alpha x} \text{Hh}_n(\beta x - \delta) \rightarrow 0, \text{ as } x \rightarrow \infty, \quad (34)$$

for any constant $\alpha < 0$, thanks to (32). Using integration by parts and (34), we again have a recursion, for $n \geq 0$,

$$I_n = -\frac{e^{\alpha \lambda}}{\alpha} \text{Hh}_n(\beta \lambda - \delta) + \frac{\beta}{\alpha} I_{n-1},$$

but with a different initial condition

$$I_{-1} = \sqrt{2\pi} \int_{\lambda}^{\infty} e^{\alpha x} \varphi(-\beta x + \delta) dx = -\frac{\sqrt{2\pi}}{\beta} \exp\left\{\frac{\alpha \delta}{\beta} + \frac{\alpha^2}{2\beta^2}\right\} \Phi\left(\beta \lambda - \delta - \frac{\alpha}{\beta}\right).$$

Solving it yields, for $n \geq -1$,

$$I_n = -\frac{e^{\alpha \lambda}}{\alpha} \sum_{i=0}^n \left(\frac{\beta}{\alpha}\right)^{n-i} \text{Hh}_i(\beta \lambda - \delta) - \left(\frac{\beta}{\alpha}\right)^{n+1} \frac{\sqrt{2\pi}}{\beta} \exp\left\{\frac{\alpha \delta}{\beta} + \frac{\alpha^2}{2\beta^2}\right\} \Phi\left(\beta \lambda - \delta - \frac{\alpha}{\beta}\right),$$

where the sum over an empty set is again defined to be zero. $\square$

B.3 Sum of double exponential and the normal random variables

Proposition B.3. Suppose $\{\xi_1, \xi_2, \dots\}$ is a sequence of i.i.d. exponential random variables with rate $\eta > 0$, and Z is a normal random variable with distribution $N(0, \sigma^2)$. Then for every $n \geq 1$, we have:

1. The density functions are given by

$$f_{Z+\sum_{i=1}^n \xi_i}(t) = (\sigma\eta)^n \frac{e^{(\sigma\eta)^2/2}}{\sigma\sqrt{2\pi}} e^{-t\eta} \text{Hh}_{n-1}\left(-\frac{t}{\sigma} + \sigma\eta\right), \quad (35)$$

$$f_{Z-\sum_{i=1}^n \xi_i}(t) = (\sigma\eta)^n \frac{e^{(\sigma\eta)^2/2}}{\sigma\sqrt{2\pi}} e^{t\eta} \text{Hh}_{n-1}\left(\frac{t}{\sigma} + \sigma\eta\right). \quad (36)$$

2. The tail probabilities are given by

$$\mathbb{P}(Z + \sum_{i=1}^n \xi_i \geq x) = \frac{(\sigma\eta)^n}{\sigma\sqrt{2\pi}} e^{(\sigma\eta)^2/2} I_{n-1}(x; -\eta, -\frac{1}{\sigma}, -\sigma\eta), \quad (37)$$

$$\mathbb{P}(Z - \sum_{i=1}^n \xi_i \geq x) = \frac{(\sigma\eta)^n}{\sigma\sqrt{2\pi}} e^{(\sigma\eta)^2/2} I_{n-1}(x; \eta, \frac{1}{\sigma}, -\sigma\eta). \quad (38)$$

3. For $n \geq 1$ and $c < \eta$,

$$\begin{aligned} & \mathbb{E}(ae^{b+c(Z+\sum_{i=1}^n \xi_i)} - K)^+ \\ &= ae^b \frac{(\sigma\eta)^n}{\sigma\sqrt{2\pi}} e^{(\sigma\eta)^2/2} I_{n-1}(h; c - \eta, -\frac{1}{\sigma}, -\sigma\eta) - K \frac{(\sigma\eta)^n}{\sigma\sqrt{2\pi}} e^{(\sigma\eta)^2/2} I_{n-1}(h; -\eta, -\frac{1}{\sigma}, -\sigma\eta). \end{aligned} \quad (39)$$

For all $n \geq 1$ and $\eta > -c$,

$$\begin{aligned} & \mathbb{E}(ae^{b+c(Z-\sum_{i=1}^n \xi_i)} - K)^+ \\ &= ae^b \frac{(\sigma\eta)^n}{\sigma\sqrt{2\pi}} e^{(\sigma\eta)^2/2} I_{n-1}(h; c + \eta, \frac{1}{\sigma}, -\sigma\eta) - K \frac{(\sigma\eta)^n}{\sigma\sqrt{2\pi}} e^{(\sigma\eta)^2/2} I_{n-1}(h; \eta, \frac{1}{\sigma}, -\sigma\eta), \end{aligned} \quad (40)$$

where $h = \frac{\log(K/a) - b}{c}$.

Proof. Case 1: the densities of $Z + \sum_{i=1}^n \xi_i$ and $Z - \sum_{i=1}^n \xi_i$. We have

$$\begin{aligned} f_{Z+\sum_{i=1}^n \xi_i}(t) &= \int_{-\infty}^{\infty} f_{\sum_{i=1}^n \xi_i}(t-x) f_Z(x) dx \\ &= e^{-t\eta} (\eta^n) \int_{-\infty}^t \frac{e^{x\eta} (t-x)^{n-1}}{(n-1)!} \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/(2\sigma^2)} dx \\ &= e^{-t\eta} (\eta^n) e^{(\sigma\eta)^2/2} \int_{-\infty}^t \frac{(t-x)^{n-1}}{(n-1)!} \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\sigma^2\eta)^2/(2\sigma^2)} dx. \end{aligned}$$

Letting $y = (x - \sigma^2\eta) / \sigma$ yields

$$\begin{aligned} f_{Z+\sum_{i=1}^n \xi_i}(t) &= e^{-t\eta} e^{(\sigma\eta)^2/2} \sigma^{n-1} \eta^n \int_{-\infty}^{t/\sigma - \sigma\eta} \frac{(t/\sigma - y - \sigma\eta)^{n-1}}{(n-1)!} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy \\ &= \frac{e^{(\sigma\eta)^2/2}}{\sqrt{2\pi}} (\sigma^{n-1} \eta^n) e^{-t\eta} \text{Hh}_{n-1}(-t/\sigma + \sigma\eta), \end{aligned}$$

because $\frac{1}{(n-1)!} \int_{-\infty}^a (a-y)^{n-1} e^{-y^2/2} dy = \text{Hh}_{n-1}(-a)$. Similarly,

$$\begin{aligned} f_{Z-\sum_{i=1}^n \xi_i}(t) &= e^{t\eta} \eta^n \int_t^{\infty} \frac{e^{-x\eta} (x-t)^{n-1}}{(n-1)!} \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/(2\sigma^2)} dx \\ &= e^{t\eta} \eta^n e^{(\sigma\eta)^2/2} \int_t^{\infty} \frac{(x-t)^{n-1}}{(n-1)!} \frac{1}{\sigma\sqrt{2\pi}} e^{-(x+\sigma^2\eta)^2/(2\sigma^2)} dx. \end{aligned}$$

Letting $\tilde{y} = (x + \sigma^2\eta) / \sigma$ leads to

$$\begin{aligned} f_{Z-\sum_{i=1}^n \xi_i}(t) &= e^{t\eta} e^{(\sigma\eta)^2/2} \sigma^{n-1} \eta^n \int_{t/\sigma + \sigma\eta}^{\infty} \frac{(\tilde{y} - \sigma\eta - t/\sigma)^{n-1}}{(n-1)!} \frac{1}{\sqrt{2\pi}} e^{-\tilde{y}^2/2} d\tilde{y} \\ &= \frac{e^{(\sigma\eta)^2/2}}{\sqrt{2\pi}} (\sigma^{n-1} \eta^n) e^{t\eta} \text{Hh}_{n-1}(t/\sigma + \sigma\eta), \end{aligned}$$

and the proof of Case 1 is terminated.

Case 2: $\mathbb{P}(Z + \sum_{i=1}^n \xi_i \geq x)$ and $\mathbb{P}(Z - \sum_{i=1}^n \xi_i \geq x)$. From the density given in (35), it is clear that

$$\mathbb{P}(Z + \sum_{i=1}^n \xi_i \geq x) = \frac{(\sigma\eta)^n e^{(\sigma\eta)^2/2}}{\sigma\sqrt{2\pi}} \int_x^{\infty} e^{-t\eta} \text{Hh}_{n-1}\left(-\frac{t}{\sigma} + \sigma\eta\right) dt = \frac{(\sigma\eta)^n e^{(\sigma\eta)^2/2}}{\sigma\sqrt{2\pi}} I_{n-1}(x; -\eta, -\frac{1}{\sigma}, -\sigma\eta),$$

by (22) and (24). Similarly, the density in (36) leads to

$$\mathbb{P}(Z - \sum_{i=1}^n \xi_i \geq x) = \frac{(\sigma\eta)^n e^{(\sigma\eta)^2/2}}{\sigma\sqrt{2\pi}} \int_x^{\infty} e^{t\eta} \text{Hh}_{n-1}\left(\frac{t}{\sigma} + \sigma\eta\right) dt = \frac{(\sigma\eta)^n e^{(\sigma\eta)^2/2}}{\sigma\sqrt{2\pi}} I_{n-1}(x; \eta, \frac{1}{\sigma}, -\sigma\eta),$$

by (22) and (23).

Case 3: $\mathbb{E}(ae^{b+c(Z+\sum_{i=1}^n \xi_i)} - K)^+$ and $\mathbb{E}(ae^{b+c(Z-\sum_{i=1}^n \xi_i)} - K)^+$. Using the density given in (35) yields, for $n \geq 1$,

$$\begin{aligned} &\mathbb{E}(ae^{b+c(Z+\sum_{i=1}^n \xi_i)} - K)^+ \\ &= ae^b \int_h^{\infty} e^{ct} f_{Z+\sum_{i=1}^n \xi_i}(t) dt - K \mathbb{P}(Z + \sum_{i=1}^n \xi_i \geq h) \\ &= ae^b (\sigma\eta)^n \frac{e^{\sigma^2/(2\eta^2)}}{\sigma\sqrt{2\pi}} \int_h^{\infty} e^{ct} e^{-t\eta} \text{Hh}_{n-1}\left(-\frac{t}{\sigma} + \sigma\eta\right) dt - K \mathbb{P}(Z + \sum_{i=1}^n \xi_i \geq h) \\ &= ae^b (\sigma\eta)^n \frac{e^{(\sigma\eta)^2/2}}{\sigma\sqrt{2\pi}} I(h; c - \eta, -\frac{1}{\sigma}, -\sigma\eta) - K \frac{(\sigma\eta)^n}{\sigma\sqrt{2\pi}} e^{(\sigma\eta)^2/2} I_{n-1}(x; -\eta, -\frac{1}{\sigma}, -\sigma\eta), \end{aligned}$$

by (24) and (37), where $h = \{\log(K/a) - b\}/c$. Similarly, using the density given in (36) yields, for $n \geq 1$,

$$\begin{aligned}
& \mathbb{E}(ae^{b+c(Z-\sum_{i=1}^n \xi_i)} - K)^+ \\
&= ae^b \int_h^\infty e^{ct} f_{Z-\sum_{i=1}^n \xi_i}(t) dt - K\mathbb{P}(Z - \sum_{i=1}^n \xi_i \geq h) \\
&= ae^b (\sigma\eta)^n \frac{e^{\sigma^2/(2\eta^2)}}{\sigma\sqrt{2\pi}} \int_h^\infty e^{ct} e^{t\eta} \text{Hh}_{n-1}\left(\frac{t}{\sigma} + \sigma\eta\right) dt - K\mathbb{P}(Z - \sum_{i=1}^n \xi_i \geq h) \\
&= ae^b (\sigma\eta)^n \frac{e^{(\sigma\eta)^2/2}}{\sigma\sqrt{2\pi}} I_{n-1}(h; c + \eta, \frac{1}{\sigma}, -\sigma\eta) - K \frac{(\sigma\eta)^n}{\sigma\sqrt{2\pi}} e^{(\sigma\eta)^2/2} I_{n-1}(x; \eta, \frac{1}{\sigma}, -\sigma\eta).
\end{aligned}$$

by (23) and (38). The proof is terminated. $\square$

B.4 Proofs of Theorem 2 and Corollary 3

Proof of Theorem 2. Conditioning on $N(t)$ leads to, via independence,

$$\psi_c(0) = e^{-rT} \sum_{n=0}^\infty \pi_n \mathbb{E}^* \left(S(0) e^{-\lambda\zeta T} \exp \left\{ \left(r - \frac{\sigma^2}{2} \right) T + \sigma\sqrt{T}Z + \sum_{j=1}^n Y_j \right\} - K \right)^+,$$

where recalling that $V_j = e^{X_i}$ and $\prod$ over the empty set is set to be one. By the decomposition (29),

$$\begin{aligned}
\psi_c(0) &= e^{-rT} \sum_{n=1}^\infty \pi_n \sum_{k=1}^n P_{n,k} \mathbb{E}^* \left(S(0) e^{-\lambda\zeta T} \exp \left\{ \left(r - \frac{\sigma^2}{2} \right) T + \sigma\sqrt{T}Z + \sum_{j=1}^k \xi_j^+ \right\} - K \right)^+ \\
&\quad + e^{-rT} \sum_{n=1}^\infty \pi_n \sum_{k=1}^n Q_{n,k} \mathbb{E}^* \left(S(0) e^{-\lambda\zeta T} \exp \left\{ \left(r - \frac{\sigma^2}{2} \right) T + \sigma\sqrt{T}Z - \sum_{j=1}^k \xi_j^- \right\} - K \right)^+ \\
&\quad + \pi_0 e^{-rT} \mathbb{E}^* \left(S(0) e^{-\lambda\zeta T} \exp \left\{ \left(r - \frac{\sigma^2}{2} \right) T + \sigma\sqrt{T}Z \right\} - K \right)^+.
\end{aligned}$$

Now substituting $a = S(0) e^{-\lambda\zeta T}$, $b = \left(r - \frac{\sigma^2}{2} \right) T$, and $c = 1$ into (39) and (40) gives, for $\eta_1 > 1$ and $\eta_2 > 0$.

$$\begin{aligned}
& \mathbb{E}^* \left(S(0) e^{-\lambda\zeta T} \exp \left\{ \left(r - \frac{\sigma^2}{2} \right) T + \sigma\sqrt{T}Z + \sum_{j=1}^k \xi_j^+ \right\} - K \right)^+ \\
&= S(0) e^{-\lambda\zeta T} e^{\left(r - \frac{\sigma^2}{2} \right) T} \frac{(\sigma\sqrt{T}\eta_1)^n}{\sigma\sqrt{T}\sqrt{2\pi}} e^{(\sigma\eta_1)^2 T/2} I_{n-1}(h; 1 - \eta_1, -\frac{1}{\sigma\sqrt{T}}, -\sigma\eta_1\sqrt{T}) \\
&\quad - K \frac{(\sigma\eta_1\sqrt{T})^n}{\sigma\sqrt{T}\sqrt{2\pi}} e^{(\sigma\eta_1)^2 T/2} I_{n-1}(h; -\eta_1, -\frac{1}{\sigma\sqrt{T}}, -\sigma\eta_1\sqrt{T}),
\end{aligned}$$

$$\begin{aligned}
& \mathbb{E}^* \left(S(0) e^{-\lambda \zeta T} \exp \left\{ \left(r - \frac{\sigma^2}{2} \right) T + \sigma \sqrt{T} Z - \sum_{j=1}^k \xi_j^- \right\} - K \right)^+ \\
&= S(0) e^{-\lambda \zeta T} e^{\left(r - \frac{\sigma^2}{2} \right) T} \frac{(\sigma \sqrt{T} \eta_2)^n}{\sigma \sqrt{T} \sqrt{2\pi}} e^{(\sigma \eta_2)^2 T/2} I_{n-1}(h; 1 + \eta_2, \frac{1}{\sigma \sqrt{T}}, -\sigma \eta_2 \sqrt{T}) \\
&\quad - K \frac{(\sigma \eta_2 \sqrt{T})^n}{\sigma \sqrt{T} \sqrt{2\pi}} e^{(\sigma \eta_2)^2 T/2} I_{n-1}(h; \eta_2, \frac{1}{\sigma \sqrt{T}}, -\sigma \eta_2 \sqrt{T}),
\end{aligned}$$

where

$$h = \frac{\log(K/a) - b}{c} = \log(K/S(0)) + \lambda \zeta T - \left(r - \frac{\sigma^2}{2} \right) T.$$

The result follows by noting that

$$\mathbb{E}^* \left(S(0) \exp \left\{ \left(r - \frac{\sigma^2}{2} - \lambda \zeta \right) T + \sigma \sqrt{T} Z \right\} - K \right)^+ = S(0) e^{(r-\lambda\zeta)T} \Phi(b_+) - K \Phi(b_-).$$

Proof of Corollary 3. I shall only prove the case for the European call option, as the case of European futures options can be dealt similarly. One can verify by simple calculus that as $\eta_1 \rightarrow \infty$ and $\eta_2 \rightarrow \infty$, it is true that $\zeta \rightarrow 0$ and

$$\begin{aligned}
a_1(\sigma \sqrt{T} \eta_1)^k I_{k-1}(h; 1 - \eta_1, -\frac{1}{\sigma \sqrt{T}}, -\sigma \eta_1 \sqrt{T}) &\rightarrow \Phi(-\frac{1}{\sigma \sqrt{T}} h + \sigma \sqrt{T}), \\
d_1(\sigma \sqrt{T} \eta_1)^k I_{k-1}(h; -\eta_1, -\frac{1}{\sigma \sqrt{T}}, -\sigma \eta_1 \sqrt{T}) &\rightarrow \Phi(-\frac{1}{\sigma \sqrt{T}} h), \\
a_2(\sigma \sqrt{T} \eta_2)^k I_{k-1}(h; 1 + \eta_2, \frac{1}{\sigma \sqrt{T}}, -\sigma \eta_2 \sqrt{T}) &\rightarrow \Phi(-\frac{1}{\sigma \sqrt{T}} h + \sigma \sqrt{T}), \\
d_2(\sigma \sqrt{T} \eta_2)^k I_{k-1}(h; \eta_2, \frac{1}{\sigma \sqrt{T}}, -\sigma \eta_2 \sqrt{T}) &\rightarrow \Phi(-\frac{1}{\sigma \sqrt{T}} h).
\end{aligned}$$

By the dominated convergence theorem (the theorem is applicable as $\sum_{k=1}^n P_{n,k} + Q_{n,k} = 1$),

$$\begin{aligned}
\psi_c(0) &\rightarrow S(0) \sum_{n=1}^{\infty} \sum_{k=1}^n \pi_n(P_{n,k} + Q_{n,k}) \Phi(-\frac{1}{\sigma \sqrt{T}} h + \sigma \sqrt{T}) \\
&\quad - K e^{-rT} \sum_{n=1}^{\infty} \sum_{k=1}^n \pi_n(P_{n,k} + Q_{n,k}) \Phi(-\frac{1}{\sigma \sqrt{T}} h) + \pi_0 \{ S(0) \Phi(b_+) - K e^{-rT} \Phi(b_-) \} \\
&= S(0) (1 - \pi_0) \Phi(-\frac{1}{\sigma \sqrt{T}} h + \sigma \sqrt{T}) - K e^{-rT} (1 - \pi_0) \Phi(-\frac{1}{\sigma \sqrt{T}} h) \\
&\quad + \pi_0 \{ S(0) \Phi(b_+) - K e^{-rT} \Phi(b_-) \} \\
&= S(0) \Phi(b'_+) - e^{-rT} K \Phi(b'_-),
\end{aligned}$$

where the identity $\sum_{k=1}^n P_{n,k} + Q_{n,k} = 1$ is used, which has been proved at the end of the proof of Proposition B.1.

If $\lambda = 0$, then $\pi_n = 0$ for all $n \geq 1$, and $\pi_0 = 1$. Thus, only the last term in the pricing formula is nonzero, and $\psi_c(0) = S(0) \Phi(b'_+) - K e^{-rT} \Phi(b'_-)$. $\square$

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